

Cloud Deployment Models

Task 1

Pick any two apps you use daily (like Zomato, Instagram, or Paytm) and identify which cloud deployment model (Public, Private, Hybrid, or Community) would best suit each app if you were the CTO. Justify your choice with two reasons for each.

1. Zomato — Public Cloud

As the CTO of Zomato, I would choose a Public Cloud deployment model for two main reasons. First, Zomato's traffic is highly variable — it sees sudden spikes during meal times, festivals, and offers, so a public cloud allows the platform to scale servers up or down instantly without owning any physical hardware. Second, a public cloud works out more cost-effective for a food-delivery app like Zomato, since it only pays for the computing resources it actually uses, instead of investing heavily in maintaining its own data centres.

2. Paytm — Hybrid Cloud

As the CTO of Paytm, I would choose a Hybrid Cloud deployment model. First, since Paytm handles sensitive financial data such as bank details, UPI transactions, and wallet balances, it needs a private, tightly controlled environment to meet strict RBI security and compliance requirements for the payment-related parts of the system. Second, for less sensitive functions such as offers, notifications, and general app content, a public cloud component allows Paytm to scale cheaply and quickly during high-traffic events like cashback sales, which a hybrid setup makes possible by combining both environments.

Task 2

Create a comparison table listing the pros and cons of Public, Private, Hybrid, and Community cloud deployment models. Include at least three points each for pros and cons.

Model	Pros	Cons
Public	● Low upfront cost — no need to buy hardware. ● Highly scalable; resources can be added instantly. ● Maintained fully by the provider, so less management effort.	● Less control over the underlying infrastructure. ● Shared resources can raise security and privacy concerns. ● Performance can be affected by other tenants on the same servers.
Private	● Greater control and customisation of infrastructure. ● Higher security, suitable for sensitive data. ● Better performance since resources are not shared.	● Expensive to set up and maintain. ● Scaling requires buying additional hardware, which takes time. ● Requires an in-house or dedicated IT team to manage it.
Hybrid	● Flexibility to keep sensitive data private and scale other parts publicly.	● More complex to design, integrate, and manage. ● Requires strong networking and security between environments.

	• Cost-efficient since public cloud is used only when needed. • Supports business continuity with backup across both environments.	• Can be costlier than a purely public setup if not managed well.
Community	• Cost is shared among organisations with similar needs. • Good for organisations with common compliance or regulatory requirements. • More secure than public cloud since access is limited to member organisations.	• Less flexible since infrastructure is designed for shared/common needs. • Slower decision-making as changes need agreement among members. • Still more expensive than public cloud for an individual organisation.

Task 3

Read the case study of WhatsApp's cloud infrastructure (search online for details) and summarize which deployment model they use, explaining why this model is suitable for their business needs. Hint: Focus on scalability, security, and global reach.

Based on publicly available information about WhatsApp's infrastructure, WhatsApp mainly follows a Hybrid Cloud deployment model. The core messaging system, which handles extremely sensitive personal chats, contacts, and end-to-end encrypted data for billions of users, runs on tightly controlled, privately managed servers owned by Meta. This gives WhatsApp full control over security, encryption, and compliance, which is essential given how sensitive user conversations are.

At the same time, WhatsApp relies on public-cloud-style components for parts of the system that need to scale quickly and reach users across the world. For example, media files (images, videos, and documents) are temporarily stored using distributed object storage and delivered through Content Delivery Networks (CDNs), which help the app load media faster no matter where a user is located. This mix allows WhatsApp to keep the core chat data secure and private while still benefiting from the elasticity and global reach of public-cloud-style infrastructure for lighter, less sensitive workloads.

This hybrid approach suits WhatsApp's business needs well: it gives them the scalability to serve billions of daily active users worldwide, the security and control needed to protect private conversations, and the global reach required to deliver messages and media with low latency in every region they operate in.

Task 4

Imagine you are launching a new online ticket booking platform for IPL matches. Choose a cloud deployment model for your platform, list two possible risks with your choice, and suggest a way to mitigate one of those risks.

Deployment model chosen: Public Cloud

I would choose a Public Cloud deployment model for the IPL ticket booking platform. IPL matches create extremely sudden and massive spikes in traffic the moment ticket sales open, and a public cloud provider can automatically scale up servers to handle millions of users trying to book tickets within minutes, then scale back down once the

rush is over. It is also the most cost-effective option, since the platform does not need to invest in and maintain hardware that would sit mostly idle between matches.

Two possible risks:

- Since resources on a public cloud are shared with other organisations (multi-tenancy), there is a risk of data breaches or unauthorised access to users' personal and payment information if the provider's security is compromised.
- The platform becomes dependent on a single cloud provider's uptime and pricing (vendor lock-in); if the provider faces an outage during a high-demand ticket sale, or suddenly raises prices, it could directly affect bookings and revenue.

Mitigation for the data security risk:

To reduce the risk of data breaches, the platform can encrypt all sensitive data (such as payment details and personal information) both at rest and in transit, enforce strict access controls and multi-factor authentication for internal systems, and regularly run security audits and penetration tests on the cloud environment. This ensures that even though the infrastructure is shared, the actual user data remains protected.